

Software Requirements Specification (SRS) - Team 3

Project: Student Attendance Management System (SAMS)

Version: 1.0

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Revision History

Version	Date	Author	Change Summary	Approval
1.0	04-09-2026	Priyanka, Priyanka M Reddy, Priyanka N, Priyanka Sajjan	Initial SRS with diagrams embedded	

Approvals

Role	Name	Signature / Email / Date
Course Coordinator		
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the Student Attendance Management System (SAMS). The purpose of this document is to provide a clear and complete description of the expected behavior, features, interfaces, constraints, and quality requirements of the system.

The SRS serves as a reference for the project team during design, development, testing, and integration. It also provides a basis for verifying that the completed SAMS satisfies the requirements defined for the project.

The system is designed as a role-based application in which an Admin manages courses and accounts, a Teacher marks and manages attendance for assigned sections, and a Student views their own attendance records and submits leave requests.

1.2 Scope

SAMS is a role-based application that allows an educational institution to record, manage, and analyze student attendance across courses and sections. The system provides distinct views and permissions for Admin, Teacher, and Student users.

The main functions within the scope of the system include:

- Authenticating users and enforcing role-based access control.
- Allowing the Admin to manage user accounts, courses, sections, and schedules.
- Allowing the Admin to enroll, transfer, or remove students from sections.
- Allowing a teacher to mark attendance for each scheduled session.
- Allowing a teacher to correct attendance records within a defined window.
- Calculating and displaying attendance percentages per course and overall.
- Generating a defaulter list based on a configurable attendance threshold.
- Allow Admin and Teacher users to export attendance reports.
- Sending low-attendance alert notifications to students.
- Allowing students to submit leave requests for Teacher approval.

The project does not include biometric or hardware-based attendance capture (e.g., fingerprint or RFID devices), payroll integration, or mobile push-notification infrastructure. The system is intended to operate as a web-based application accessed through a standard browser. This is consistent with the defined project scope for a role-based Student Attendance Management System.

1.3 Audience

This document is intended for:

- Project Developers: To understand the required functionality and system behavior during implementation.
- Testers/QA Members: To create and execute test cases based on the defined acceptance criteria.
- Project Team Members: To coordinate the development of different system features and ensure that the modules work together correctly.
- Course Coordinator/Instructor: To evaluate whether the project satisfies the specified software requirements.
- Assessment of Evaluators: To verify the functional and non-functional requirements of the completed system.

1.4 Definitions

The following terms are used throughout this document:

Term	Definition
Student	An enrolled user whose class attendance is recorded and who can view their own attendance records.
Teacher	A staff user responsible for marking and correcting attendance for the sections/courses assigned to them.
Admin	A user role responsible for managing accounts, courses, sections, and system configuration.
Attendance Record	A single entry capturing a student's presence status for a specific session, date, and course.
Session / Period	A scheduled class meeting (lecture, lab, or period) for which attendance is taken.
Attendance Percentage	The proportion of sessions a student was marked Present out of the total sessions held, expressed as a percentage.
Defaulter	A student whose attendance percentage falls below the configured minimum threshold.
Correction Window	The permitted time period during which a submitted attendance record may be edited.
Roster	The list of students enrolled in a given course or section.
Functional Requirement (FR)	A requirement describing what the system must do.
Non-Functional Requirement (NFR)	A requirement describing qualities or constraints of the system, such as performance and usability.
UML	Unified Modeling Language used to represent system design and behavior.
RTM	Requirements Traceability Matrix used to map requirements to their corresponding test cases.

2. Overall description

SAMS is a role-based, browser-accessible application. The system does not depend on hardware attendance-capture devices and operates as a standalone web application within the institution's network or hosting environment.

The system consists of the following main components:

- User Interface - provides role-specific dashboards for Admin, Teacher, and Student.
- Authentication and Access Control - manages login, sessions, and role-based permissions.
- Course and Enrollment Management - maintains courses, sections, schedules, and student rosters.
- Attendance Engine - processes attendance marking, edits, and percentage calculations.
- Reporting and Notification - generates reports, defaulter lists, exports, and alerts.

2.2 Major Product Functions

The major functions of SAMS are:

1. Authentication and Role Management - secure login and role-based access for Admin, Teacher, and Student.
2. Course and Class Management - creation and administration of courses, sections, and schedules.
3. Student Enrollment Management - maintaining accurate class rosters.
4. Attendance Marking and Recording - capturing and correcting per-session attendance.
5. Attendance Reports and Analytics - percentage calculation, defaulter lists, and exports.
6. Notifications and Leave Management - low-attendance alerts and leave request handling.

2.3 User Roles and Characteristics

The primary users of the system are the Admin, Teacher, and Student.

Admin

The Admin manages user accounts, courses, sections, schedules, and system-wide configuration such as the attendance threshold. The Admin has access to all reports and audit information.

Teacher

The Teacher marks and corrects attendance for assigned sections, views reports for their own sections, and reviews student leave requests.

Student

The Student views their own attendance records and percentage, receives low-attendance alerts, and submits leave requests. No specialized technical knowledge is required to use the system.

2.4 Operating Environment

SAMS is intended to run in the following environment:

- A modern web browser on a desktop, laptop, or tablet device.
- A minimum display resolution of 1024 x 768 pixels.
- A stable internet or institutional network connection.
- A supported web server and database backend hosting the application.
- Standard keyboard and mouse/touch input.

The system requires a network connection to the hosting server and is not designed to operate fully offline.

2.5 Constraints

SAMS is subject to the following constraints:

- The system supports three user roles: Admin, Teacher, and Student, each with distinct permissions.
- Attendance data must be preserved even when a student is transferred or removed from a section.
- Attendance edits are only permitted within a defined correction window.
- The system must remain responsive under expected institutional load (see Section 5).
- The implementation is dependent on the selected web framework, database, and hosting environment.
- The user interface and workflow behavior should remain consistent with the functional and non-functional requirements specified in this SRS.

3. External interface requirements

3.1 User Interfaces

The system presents role-specific dashboards: an Admin console (user, course, and configuration management), a Teacher console (roster, attendance marking, corrections, leave review), and a Student console (attendance percentage, notifications, leave submission). All screens use a single consistent color/typography scheme and provide clear navigation and status indicators appropriate to each role.

3.2 Hardware Interfaces

Keyboard and mouse/touch: used for navigation, form entry, and selection across all role dashboards.

Display: standard monitor, laptop, or tablet screen, minimum resolution assumed 1024x768.

3.3 Software Interfaces

Web application server and database used to store and retrieve user, course, enrollment, and attendance data.

Notification/email service interface used to deliver low-attendance alerts and leave-request status updates.

3.4 Communications

SAMS communicates over standard HTTPS between the browser client and the application server. No third-party hardware attendance-capture devices are included in this project's scope.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case.

4.1 User Authentication and Role Management

On launch, a user logs in with their ID and password. The system authenticates the user and routes them to the dashboard appropriate to their role (Admin, Teacher, or Student), enforcing role-based access to every subsequent screen and action described in Sections 4.2-4.6.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAMS-F-001	The system shall allow Admin, Teacher, and Student users to log in using a unique ID and password.	Functional	High	Admin / Teacher / Student	AC-SAMS-F-001: On providing valid credentials, the user is authenticated and redirected to their role-specific dashboard within 2s. Test: TC-Auth-01	Foundational; all other features require an authenticated session.
SAMS-F-002	The system shall enforce role-based access control, restricting each user to the screens and actions permitted for their role.	Functional	High	Admin / Teacher / Student	AC-SAMS-F-002: A Teacher attempting to access Admin-only screens is denied access and redirected. Test: TC-Auth-02	Depends on successful login (SAMS-F-001).
SAMS-F-003	The system shall allow the Admin to create, edit, and deactivate Teacher and Student user accounts.	Functional	High	Admin	AC-SAMS-F-003: A newly created account can log in immediately; a deactivated account is denied login. Test: TC-Auth-03	Deactivation must not delete historical attendance data.

4.2 Course and Class Management

Description: Create and administer the courses, sections, and schedules that all subsequent enrollment and attendance activity depends on.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
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Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAMS-F-004	The system shall allow the Admin to create and manage courses, sections, and class schedules.	Functional	High	Admin	AC-SAMS-F-004: A newly created course/section appears in the course list and is available for teacher assignment and enrollment. Test: TC-Course-01	Required before enrollment or attendance marking can occur.
SAMS-F-005	The system shall allow the Admin to assign one or more Teachers to each course/section.	Functional	Medium	Admin	AC-SAMS-F-005: The assigned Teacher can see the section in their dashboard and mark attendance for it. Test: TC-Course-02	Depends on SAMS-F-004.

4.3 Student Enrollment Management

Description: Maintain accurate, up-to-date class rosters used by attendance marking, while preserving historical data.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAMS-F-006	The system shall allow the Admin to enroll students into a course/section and maintain an up-to-date class roster.	Functional	High	Admin	AC-SAMS-F-006: An enrolled student appears on the roster used for attendance marking within the same session. Test: TC-Enroll-01	Roster accuracy directly affects attendance marking (Section 4.4).
SAMS-F-007	The system shall allow the Admin to transfer or remove a student from a section while preserving that student's historical attendance records.	Functional	Medium	Admin	AC-SAMS-F-007: After removal/transfer, prior attendance entries for the student remain queryable in reports. Test: TC-Enroll-02	Historical data must not be deleted on roster change.

4.4 Attendance Marking and Recording

Description: Capture per-session attendance for each enrolled student and allow controlled corrections within a defined time window.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAMS-F-008	The system shall allow a Teacher to mark attendance (Present / Absent / Late) for each enrolled student during a scheduled class session.	Functional	High	Teacher	AC-SAMS-F-008: For a section of up to 60 students, the Teacher can mark and submit attendance for every roster entry in a single session. Test: TC-Mark-01	Depends on an initialized roster (SAMS-F-006).
SAMS-F-009	The system shall record the date, session/period, and the identity of the Teacher who marked each attendance entry.	Functional	High	Teacher / System	AC-SAMS-F-009: Every stored attendance record includes date, period, course/section, student ID, status, and marking Teacher ID. Test: TC-Mark-02	Required for audit and RTM traceability.
SAMS-F-010	The system shall allow a Teacher to edit a submitted attendance record within a defined correction window (e.g., same day), logging the change as an edit.	Functional	Medium	Teacher	AC-SAMS-F-010: An edit made within the correction window updates the record and creates an audit entry; edits attempted after the window are rejected. Test: TC-Mark-03	Depends on SAMS-F-008; see SAMS-SR-002.

4.5 Attendance Reports and Analytics

Description: Calculate attendance percentages, identify defaulters against a configurable threshold, and support report export.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAMS-F-011	The system shall calculate and display each student's attendance percentage per course and overall, across all enrolled courses.	Functional	High	Student / Teacher / Admin	AC-SAMS-F-011: The displayed percentage equals (sessions present / total sessions held) x 100, recalculated after every new attendance submission. Test: TC-Report-01	Depends on accumulated attendance records (Section 4.4).
SAMS-F-012	The system shall generate a defaulter list of students whose attendance percentage falls below a configurable threshold.	Functional	High	Admin / Teacher	AC-SAMS-F-012: With the threshold set to 75%, only students below 75% attendance appear on the generated list for the selected course/section. Test: TC-Report-02	Threshold is configurable per SAMS-F-014's alerting logic.
SAMS-F-013	The system shall allow the Admin and Teacher to export attendance reports (CSV/PDF) for a selected course, section, and date range.	Functional	Medium	Admin / Teacher	AC-SAMS-F-013: The exported file contains attendance data matching the on-screen report for the selected filters. Test: TC-Report-03	Depends on SAMS-F-011.

4.6 Notifications and Leave Management

Description: Alert students proactively about low attendance and allow students to submit leave requests for Teacher review.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAMS-F-014	The system shall send a low-attendance alert notification to a Student when their attendance percentage falls below the configured threshold.	Functional	Medium	Student / System	AC-SAMS-F-014: When a student's attendance drops below the threshold after an attendance update, a notification is generated within the same session. Test: TC-Notify-01	Depends on SAMS-F-011 and SAMS-F-012.
SAMS-F-015	The system shall allow a Student to submit a leave/absence request with a reason, which the assigned Teacher can approve or reject.	Functional	Low	Student / Teacher	AC-SAMS-F-015: A submitted request appears in the Teacher's pending list; the Teacher's decision updates the request status visible to the Student. Test: TC-Notify-02	Approved leave does not automatically alter attendance status; Teacher confirms final marking separately.

5. Non-functional requirements

NFRs below are measurable and tied to test plans.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
SAMS-NF-001	The system shall save a submitted attendance entry for a class of up to 60 students within 3 seconds of Teacher submission.	Performance	High	AC-SAMS-NF-001: Submission-to-confirmation latency measured at <=3s in 95% of sampled submissions during a test session. Test: TC-Perf-Mark-01
SAMS-NF-002	The system shall not lose any submitted attendance data due to a network interruption or session timeout during submission.	Reliability	High	AC-SAMS-NF-002: Simulated interruption during submission results in either a confirmed save or a clear retry prompt, with no partial/duplicate records. Test: TC-Rel-Mark-01
SAMS-NF-003	The attendance marking screen shall be fully operable by a Teacher with no prior training,	Usability	Medium	AC-SAMS-NF-003: A first-time Teacher user completes attendance marking for a 40-student

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
	completing marking for a 40-student class within 2 minutes.			roster within 2 minutes in a usability walkthrough. Test: TC-UX-Mark-01
SAMS-NF-004	The system shall support at least 100 concurrent users marking or viewing attendance without noticeable performance degradation.	Performance	High	AC-SAMS-NF-004: Load test with 100 concurrent sessions shows average response time <=2s and no failed requests. Test: TC-Perf-Load-01
SAMS-NF-005	Attendance percentage and defaulter-list reports shall be generated and displayed within 5 seconds for a section of up to 200 students.	Performance	Medium	AC-SAMS-NF-005: Report generation for a 200-student section completes in <=5s in 95% of sampled test runs. Test: TC-Perf-Report-01
SAMS-NF-006	The system shall run for at least 8 continuous hours of normal school-day operation without crashing or degrading response time.	Reliability	High	AC-SAMS-NF-006: Soak test of 8+ hours completes with no crash and stable response times. Test: TC-Rel-Soak-01

5.1 Security

5.1.1 Security Objectives

Security Objective 1

SO-001: The system shall protect student attendance and personal data from unauthorized modification or deletion.

Security Objective 2

SO-002: The system shall prevent unauthorized users or processes from modifying course configuration, stored attendance records, or other persistent system information.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case Ref
SAMS-SR-001	The system shall validate all user-provided input, including login credentials and attendance data, before storing or processing it.	Security	High	AC-SAMS-SR-001: Invalid, excessively long, or unsupported input is rejected without causing a crash or corrupting stored data. Test: TC-Sec-01
SAMS-SR-002	The system shall prevent unauthorized modification of submitted attendance records, permitting edits only by the marking Teacher (within the correction window) or an Admin.	Security	High	AC-SAMS-SR-002: An edit attempt by an unauthorized user or process is rejected and the original record remains unchanged. Test: TC-Sec-02
SAMS-SR-003	The system shall restrict administrative and configuration operations (user management, threshold settings) to authorized Admin accounts.	Security	Medium	AC-SAMS-SR-003: A non-Admin attempting an administrative operation is denied; an authorized Admin can perform it successfully. Test: TC-Sec-03
SAMS-SR-004	The system shall protect stored student personal information and attendance records from unauthorized access or disclosure.	Security	Medium	AC-SAMS-SR-004: A user can access only the student records permitted for their role (own data for Students, assigned sections for Teachers, all data for Admin). Test: TC-Sec-04
SAMS-SR-005	The system shall automatically expire authenticated user sessions after 30 minutes of inactivity and require re-authentication before further access.	Security	Medium	AC-SAMS-SR-005: After 30 minutes of inactivity, the authenticated session is invalidated and access to protected screens requires the user to log in again. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

6.1 Interfaces and Basic Access acceptance

Test Case ID	Requirement ID	Test Objective	Test Procedure	Expected Result
TC-Auth-01	SAMS-F-001	Verify that valid credentials authenticate a user and route them to their role dashboard.	1. Open login page. 2. Enter valid ID and password for each role. 3. Submit. 4. Observe redirect.	User is authenticated and redirected to the correct role-specific dashboard within 2 seconds.
TC-Auth-02	SAMS-F-002	Verify role-based access control blocks out-of-role screens.	1. Log in as Teacher. 2. Attempt to navigate to an Admin-only URL/screen. 3. Observe result.	Access is denied and the user is redirected with an appropriate message.
TC-Auth-03	SAMS-F-003	Verify Admin can create and deactivate accounts, and that access reflects the change.	1. Log in as Admin. 2. Create a new Teacher account. 3. Log in as that Teacher. 4. Deactivate the account as Admin. 5. Attempt login again.	New account logs in successfully; after deactivation, login is denied.
TC-Course-01	SAMS-F-004	Verify Admin can create a course/section and schedule.	1. Log in as Admin. 2. Create a new course and section with a schedule. 3. Observe course list.	The new course/section appears in the list and is available for assignment and enrollment.
TC-Course-02	SAMS-F-005	Verify Admin can assign a Teacher to a section and the Teacher gains access.	1. Log in as Admin. 2. Assign a Teacher to a section. 3. Log in as that Teacher.	The assigned section is visible on the Teacher's dashboard and attendance marking is enabled for it.

6.2 Core Attendance Workflow acceptance

Test Case ID	Requirement ID	Test Objective	Test Procedure	Expected Result
TC-Enroll-01	SAMS-F-006	Verify enrolled students appear correctly on the class roster.	1. Log in as Admin. 2. Enroll a student into a section. 3. Log in as the assigned Teacher. 4. Open the attendance marking screen for that section.	The enrolled student appears on the roster available for attendance marking.
TC-Enroll-02	SAMS-F-007	Verify historical attendance is preserved after a student is transferred or removed.	1. Mark several attendance entries for a student. 2. Remove/transfer the student from the section as Admin. 3. Open the historical attendance report for that student.	Prior attendance entries remain visible and unchanged in the report.
TC-Mark-01	SAMS-F-008	Verify Teacher can mark attendance for an entire roster in one session.	1. Log in as Teacher. 2. Open a 60-student section. 3. Mark Present/Absent/Late for all students. 4. Submit.	All 60 entries are saved and reflected on the roster with the selected status.
TC-Mark-02	SAMS-F-009	Verify each attendance entry stores date, period, and marking Teacher.	1. Mark attendance as Teacher. 2. Inspect the stored record (via report/detail view).	The record shows correct date, session/period, course/section, and the marking Teacher's identity.
TC-Mark-03	SAMS-F-010	Verify attendance edits are accepted within the correction window and rejected afterward.	1. Mark attendance. 2. Edit the same-day record. 3. Simulate the window elapsing. 4. Attempt another edit.	The same-day edit succeeds and is logged; the later edit attempt is rejected.
TC-Report-01	SAMS-F-011	Verify attendance percentage is calculated correctly.	1. Accumulate a known mix of Present/Absent entries for a student. 2. View the student's attendance percentage.	Displayed percentage matches (sessions present / total sessions) x 100.
TC-Report-02	SAMS-F-012	Verify the defaulter list correctly filters by threshold.	1. Set threshold to 75%. 2. Generate defaulter list for a section with mixed attendance levels.	Only students below 75% attendance appear on the list.

6.3 Advanced features, Security, UML acceptance

Test Case ID	Requirement ID	Test Objective	Test Procedure	Expected Result
TC-Report-03	SAMS-F-013	Verify report export matches the on-screen data.	1. Apply course/section/date filters. 2. Export as CSV and PDF. 3. Compare exported data to the on-screen report.	Exported file content matches the filtered on-screen report exactly.
TC-Notify-01	SAMS-F-014	Verify a low-attendance alert is generated when a student drops below threshold.	1. Mark attendance so a student's percentage falls below the threshold. 2. Observe notification queue/inbox.	A notification is generated for the student within the same session.
TC-Notify-02	SAMS-F-015	Verify leave request submission and Teacher approval/rejection flow.	1. Log in as Student, submit a leave request with a reason. 2. Log in as Teacher, review and approve/reject it.	The request appears in the Teacher's pending list; the Student sees the updated status after the decision.
TC-Perf-Mark-01	SAMS-NF-001	Verify attendance submission completes within the required time.	1. Submit attendance for a 60-student section. 2. Measure time from submit to confirmation across multiple runs.	Submission confirms within 3 seconds in at least 95% of sampled runs.
TC-Sec-01	SAMS-SR-001	Verify input validation on login and attendance data.	1. Enter valid input. 2. Enter invalid, excessively long, and unsupported input into login and attendance fields. 3. Observe system behavior and stored data.	Valid input is accepted; invalid input is rejected without a crash or data corruption.
TC-Sec-02	SAMS-SR-002	Verify unauthorized attendance edits are blocked.	1. Create a valid attendance record. 2. Attempt to modify it from an unauthorized account. 3. Check the stored record.	The unauthorized edit is rejected and the original record remains unchanged.
TC-Sec-03	SAMS-SR-003	Verify administrative operations are restricted to Admin accounts.	1. Attempt an admin operation (e.g., threshold change) as Teacher. 2. Repeat as Admin.	Teacher attempt is denied; Admin attempt succeeds.
TC-Sec-04	SAMS-SR-004	Verify data access is limited to what each role is permitted to see.	1. Log in as Student and attempt to view another student's records. 2. Log in as Teacher and view only assigned sections. 3. Log in as Admin and view all data.	Each role sees only the data permitted for it; unauthorized access attempts are denied.
TC-Rel-Mark-01	SAMS-NF-002	Verify submitted attendance is not lost or duplicated when a network interruption or session timeout occurs during submission.	1. Open a 60-student attendance session. 2. Enter attendance. 3. Interrupt the network during submission or force a session interruption. 4. Restore connectivity. 5. Check the attendance records and retry behavior.	The submission is either confirmed as saved or a clear retry prompt is shown; no partial or duplicate attendance records are created.
TC-UX-Mark-01	SAMS-NF-003	Verify a first-time Teacher can complete attendance marking for a 40-student class without prior training within 2 minutes.	1. Give a first-time Teacher only the normal login instructions. 2. Open a 40-student section. 3. Mark Present/Absent/Late for all students. 4. Submit. 5. Record completion time and note usability issues.	The Teacher completes the workflow within 2 minutes without prior training, with all 40 attendance entries correctly submitted.
TC-Perf-Load-01	SAMS-NF-004	Verify the system supports at least 100 concurrent users marking or viewing attendance without noticeable performance degradation.	1. Generate 100 concurrent user sessions. 2. Distribute users across attendance marking and viewing operations. 3. Run the load test for a representative period. 4. Record response times and failed requests.	Average response time is ≤ 2 seconds and no requests fail during the 100-user load test.
TC-Perf-Report-01	SAMS-NF-005	Verify attendance percentage and defaulter-list reports are generated within 5 seconds for a section of up to 200 students.	1. Prepare a section containing 200 students with attendance records. 2. Generate the attendance percentage report and defaulter list. 3. Measure generation/display time across multiple runs.	Report generation completes within 5 seconds in at least 95% of sampled test runs.

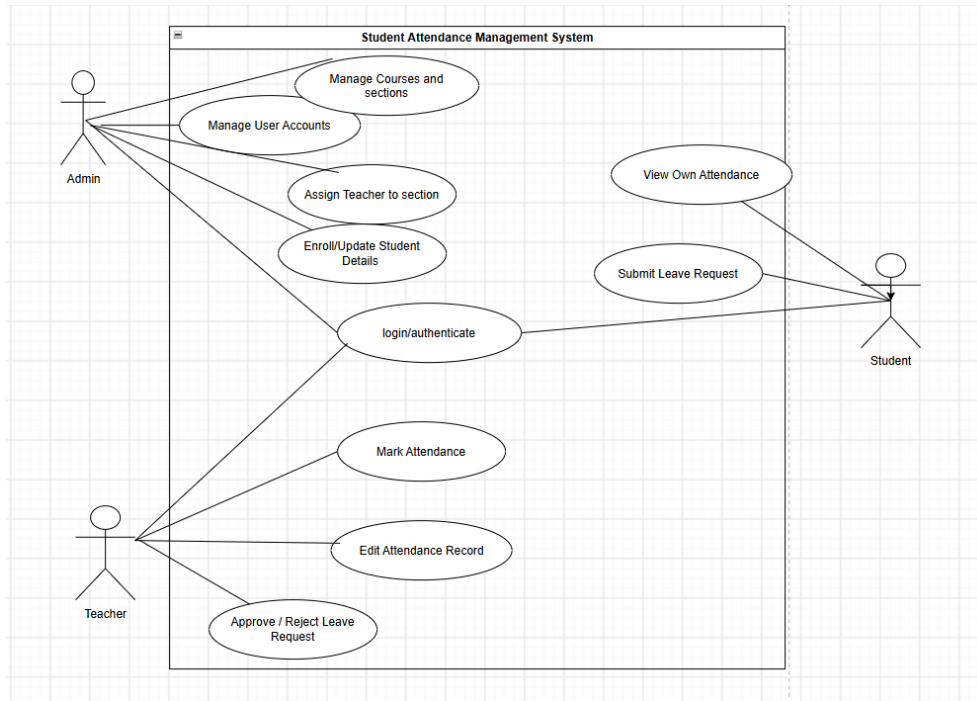
Test Case ID	Requirement ID	Test Objective	Test Procedure	Expected Result
TC-Rel-Soak-01	SAMS-NF-006	Verify the system remains stable for at least 8 continuous hours of normal school-day operation.	1. Start the system in a clean state. 2. Simulate normal attendance marking, viewing, reporting, and notification activity continuously for 8+ hours. 3. Monitor crashes, errors, and response times throughout the test. 4. Review system logs at the end.	The system completes the 8+ hour soak test without crashing, and response times remain stable with no sustained degradation.
TC-Sec-05	SAMS-SR-005	Verify inactive authenticated sessions expire and protected access requires re-authentication.	1. Log in as a valid user. 2. Leave the session inactive for 30 minutes. 3. Attempt to open a protected screen or perform a protected action. 4. Observe the result. 5. Log in again and retry the action.	The inactive session is invalidated after 30 minutes; protected access is denied until the user authenticates again.

7. System models and diagrams

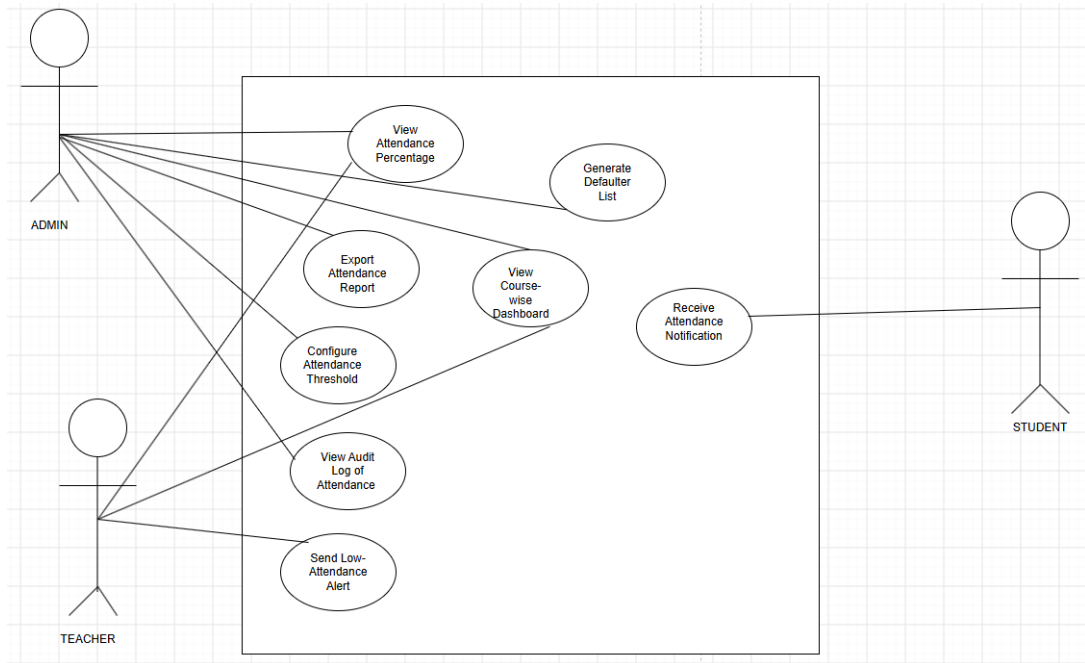
7.1 UML Use-Case diagrams

At least 2 UML use case diagrams are provided below, covering the primary Admin/Teacher/Student workflows and the reporting and notification subsystem.

UML1: Core Admin, Teacher, and Student use cases



UML2: Reporting and notification subsystem use cases



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement	Section	Module	Test Case(s)	Status (N/P/A)
SAMS-F-001	Login / authentication	4.1	AuthModule	TC-Auth-01	N
SAMS-F-002	Role-based access control	4.1	AuthModule	TC-Auth-02	N
SAMS-F-003	Account management	4.1	UserModule	TC-Auth-03	N
SAMS-F-004	Course/section management	4.2	CourseModule	TC-Course-01	N
SAMS-F-005	Teacher assignment	4.2	CourseModule	TC-Course-02	N
SAMS-F-006	Student enrollment / roster	4.3	EnrollmentModule	TC-Enroll-01	N
SAMS-F-007	Transfer / removal with history	4.3	EnrollmentModule	TC-Enroll-02	N
SAMS-F-008	Attendance marking	4.4	AttendanceModule	TC-Mark-01	N
SAMS-F-009	Attendance metadata logging	4.4	AttendanceModule	TC-Mark-02	N
SAMS-F-010	Attendance edit / correction window	4.4	AttendanceModule	TC-Mark-03	N
SAMS-F-011	Attendance percentage calculation	4.5	ReportModule	TC-Report-01	N
SAMS-F-012	Defaulter list generation	4.5	ReportModule	TC-Report-02	N
SAMS-F-013	Report export	4.5	ReportModule	TC-Report-03	N
SAMS-F-014	Low-attendance alert	4.6	NotificationModule	TC-Notify-01	N
SAMS-F-015	Leave request / approval	4.6	NotificationModule	TC-Notify-02	N
SAMS-NF-001	Attendance save latency	5	AttendanceModule	TC-Perf-Mark-01	N
SAMS-NF-002	No data loss on interruption	5	AttendanceModule	TC-Rel-Mark-01	N
SAMS-NF-003	Marking screen usability	5	UI Module	TC-UX-Mark-01	N
SAMS-NF-004	Concurrent user load	5	All modules	TC-Perf-Load-01	N
SAMS-NF-005	Report generation latency	5	ReportModule	TC-Perf-Report-01	N
SAMS-NF-006	8-hour stability	5	All modules	TC-Rel-Soak-01	N
SAMS-SR-001	Input validation	5.1.2	All modules	TC-Sec-01	N
SAMS-SR-002	Unauthorized edit prevention	5.1.2	AttendanceModule	TC-Sec-02	N
SAMS-SR-003	Admin-restricted operations	5.1.2	UserModule	TC-Sec-03	N
SAMS-SR-004	Role-based data access	5.1.2	All modules	TC-Sec-04	N
SAMS-SR-005	Session timeout / expiry	5.1.2	AuthModule	TC-Sec-05	N